

Swaroop Hegde

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Research Interests

Additive Combinatorics, Extremal Combinatorics, Probabilistic Methods in Combinatorics

Education

University of Georgia, PhD in Mathematics – Athens, GA Aug 2021 – present

- Advisor: Giorgis Petridis

National Institute of Science Education and Research (NISER), Integrated M.Sc. in Mathematics – Bhubaneswar, India Aug 2015 – May 2020

- Supervisor: Brundaban Sahu
- Master's Thesis: Combinatorics and Representation Theory of Symmetric Groups

Teaching Experience

Graduate Student Instructor, University of Georgia – Athens, GA Aug 2021 – present

Responsible for preparing course notes, delivering lectures, writing exams/quizzes, holding office hours, and grading. Courses Taught:

- **Calculus I (Math 2250)**: Spring 2025, Fall 2024, Fall 2023, Spring 2023, Fall 2022
- **PreCalculus (Math 1113)**: Spring 2026 (x2), Fall 2025 (x2), Spring 2024 (x2), Spring 2022 (x2), Fall 2021
- **Online Precalculus (Math 1113E)**: Summer 2024

Publications

An Inverse Result for Ruzsa's Inequality on Triple Sumsets Oct 2025

Swaroop Hegde

arxiv.org/abs/2510.20073 (arXiv preprint arXiv:2510.20073)

Awards

David Galewski Outstanding Graduate Teaching Award – Mathematics Department, UGA Apr 2024

Outstanding Teaching Assistant Award – Center for Teaching and Learning, UGA Mar 2024

Certificate of Merit – School of Mathematical Sciences, NISER Sept 2018

- For outstanding performance in the Spring 2018 semester.

INSPIRE Scholarship Award – Dept. of Science and Technology, Govt. of India Jan 2015

- Innovation in Science Pursuit for Inspired Research.

All-India Rank 12116 – IIT-JEE Mains Exam Jan 2015

- Top 1 percentile of all students who appeared for the exam nationwide.

Service

Co-organizer of the Graduate Student Seminar, Mathematics Department, UGA – Athens, GA Jan 2025 – May 2025

Co-facilitator of Precalculus working group, Mathematics Department, UGA – Athens, GA Jan 2024 – May 2024

Volunteer for High School Math Tournament, University of Georgia – Athens, GA Oct 2023 – Oct 2024

- Volunteered in Oct 2023 and Oct 2024.

Peer Mentor for first year graduate students, Mathematics Department, UGA – Athens, GA Aug 2023 – May 2025

Secretary, Student Chapter of AMS, University of Georgia – Athens, GA Aug 2022 – May 2023

Selected Talks

An Inverse Result for Ruzsa’s Inequality on Triple Sumsets – Discrete Math Seminar, Iowa State University (virtual) Oct 2025

An Inverse Result for Ruzsa’s Inequality on Triple Sumsets – CanaDAM Conference, Ottawa May 2025

An Inverse Result for Ruzsa’s Inequality on Triple Sumsets – Integers Conference, UGA May 2025

An Inverse Result for Ruzsa’s Inequality on Triple Sumsets – Number Theory Seminar, UGA Apr 2025

Asymptotics of the Ramsey Number $r(4,t)$ – Oral Qualifying Exam, UGA Mar 2024

AKS Algorithm for Primality Testing – Combinatorial Number Theory Seminar, UGA Oct 2023

A Polygon Problem – Graduate Student Seminar, UGA Oct 2023

The RSK Correspondence – Graduate Student Seminar, UGA Apr 2023

Almost Periods of Convolutions – Additive Combinatorics Seminar, UGA Feb 2023

Cores and Quotients of Integer Partitions – Graduate Student Seminar, UGA Nov 2022

Rédei’s Theorem and Szőnyi Method – Polynomial Methods Seminar, UGA Sept 2021

Roth’s Theorem via Finite Fourier Analysis – Math Seminar, NISER Apr 2019

Pebbling the Chessboard – NISER Open Day Apr 2018

The Story of Dimension – MathematiX Club, NISER Feb 2017

Clock Arithmetic – NISER Open Day (for high school students) Apr 2016

Gödel’s Incompleteness Theorems – MathematiX Club, NISER Nov 2015

Conferences and Workshops

PCMI 2025 Graduate Summer School on Extremal Combinatorics – Park City, Utah July 2025

Graduate Research Workshop in Combinatorics – Iowa State University, Ames, Iowa June 2025

CanaDAM Conference 2025 – Ottawa, Ontario, Canada May 2025

Integers Conference 2025 – University of Georgia, Athens, GA May 2025

Algebraic and Analytic Methods in Combinatorics – SLMATH, Berkeley, CA Mar 2025

Introductory Workshop - Graph Theory – SLMATH, Berkeley, CA Feb 2025

Early-Career Conference in Combinatorics – UIUC, Illinois June 2024

33rd Cumberland Conference on Combinatorics – Mississippi State University May 2024

Formalization of Mathematics – SLMATH, Berkeley, CA June 2023

Integers Conference 2023 – University of Georgia, Athens, GA May 2023

ATM-Schools AIS on Representation Theory of Finite Groups – CMI, Chennai, India June 2017

National Science Camp (VIJYOSHI) – Kolkata, India Dec 2015

Selected Coursework

Graduate Courses: Ramsey Theory, Additive Combinatorics I & II, Probabilistic Methods in Combinatorics, Ergodic Theory, The Circle Method, Equations with Prime Variables, Algebraic Graph Theory, Quiver Representations, Lie Algebras, Hopf Algebras

Reading Courses: Asymptotics of Ramsey Number $R(4,t)$, Kelley-Meka bound for Roth's theorem, Improvements on Freiman's Theorem, Polynomial Methods in Combinatorics, Reflection Groups

Mentoring

Directed Reading Program, University of Georgia

- Spring 2025: Mentored undergraduate student Benjamin Wang-Tie on 'Applications of Entropy in Combinatorics'.
- Summer 2021: Mentored undergraduate student Aakash Khanal on 'Mathematics of Card Shuffling'.